

Optimization of PIC codes by improved memory management

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Abstract

A simple method is described for optimization of particle-in-cell codes by improved memory management. This includes a faster calculation of Monte-Carlo collision operators. It is demonstrated that the CPU time can be reduced by a factor of 2 and more without reduction of the simulation accuracy.

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1. Introduction

Particle-in-cell (PIC) codes are extensively used for different plasma studies. Large number of simulation particles (10^5 – 10^{10}) and spatial grid cells (10^2 – 10^7) used in PIC simulations [1] make the corresponding runs extremely time consuming. Some simulation runs can take 10^4 CPU hours. As a result, the optimization of PIC codes is an important and necessary task for increasing of capabilities of corresponding simulations [2].

Originally PIC codes have been developed for simulations of collisionless plasma (see [3]) and corresponding code optimizations deal with the particle mover, the solver of the Maxwell equations and parallelization of these codes (e.g. see [4,5]). Later, different kinds of particle collisions have been added to the PIC codes, so that most of them represent hybrid PIC + Monte Carlo (MC) codes simulating a collisional plasma [6–12]. Here, under “PIC” and “MC” we mean the parts of the code simulating collisionless plasma and particle collisions, respectively. The MC part in PIC simulations can be as (or even more) time consuming as the PIC part, so that only the optimization of the PIC part is no longer sufficient.

In the present work, we describe one possible optimization, which can be easily implemented in the PIC codes and reduce the CPU time (for a single processor) up to 50%. The main idea is very simple: Simulation particle information is stored in the memory not randomly anymore, but according to the spatial grid cell. In a

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